English Language Centre

ELC6012 Thesis Writing for Research Students

Writing the Introduction chapter

I. Purposes of the Introduction chapter in a thesis

The Introduction chapter is an extremely important part of your thesis as it sets the context for the discussion in the chapters that follow. Your Introduction chapter should: i) define what is to be studied; ii) explain why it is worth studying; and iii) provide an outline of your research. In general, an introduction serves the following purposes:

- To set the context of the study by providing readers the background of your research topic;
- To situate your research in the field by showing the relationship between your work and previous studies;
- To review previous studies and identify the research gap that your study aims to address;
- To establish the significance of your research by showing how your research can address the research gap under study;
- To indicate how the study addresses the research gap being investigated;
- To indicate the questions that your research aims to answer or the research hypothesis that your study aims to test; and
- To show the outline of the thesis.

II. Moves in the Introduction chapter

The concept of "move" was introduced by Swales (1990). The term was used to describe the language features or content that often occur in academic writing. Each move represents a specific communicative act that helps fulfill the main communicative function of the entire text (Drid, 2020, p. 178). Readers rely on the moves to clarify the writer's intent. In his 'Create a Research Space' (CARS) model, Swales (1990, 2004) states that an introduction generally consists of three main moves:

- **Move 1: Establishing a territory**
- **Move 2: Establishing a niche**
- **Move 3: Occupying a niche**

Paltridge and Starfield (2007, p. 83) later modified this three-move framework and included in it the steps that writers can take to fulfill each move (Table 1).

Move 1	*Establishing a research territory* a by showing that the general research area is important, central, interesting, problematic, or relevant in some way (optional) b by providing background information about the topic (optional) c by introducing and reviewing items of previous research in the area (obligatory) d by defining terms (optional)
Move 2	*Establishing a niche* a by indicating a gap in the previous research, raising a question about it, or extending previous knowledge in some way (obligatory) b by identifying a problem/need (optional)
Move 3	*Occupying the niche* a by outlining purposes/aims, or stating the nature of the present research or research questions/hypotheses (obligatory) b by announcing principal findings/stating value of research (optional) c by indicating the structure of the thesis and providing mini-synopses (previews) of each subsequent chapter (obligatory) d by outlining the theoretical position (optional) e by describing the methods used in the study (optional)

Source: based on Swales and Feak 1994: 175 and Bunton 2002: 67

Table 1 Typical moves in thesis Introductions

As you can see from Table 1, each main move consists of a number of sub-moves. It should be noted that some sub-moves (labelled 'optional' in Table 1) may not be found in an Introduction chapter. In Move 1, writers emphasise the importance and significance of their research area by reviewing previous studies in the field. To help set the context for discussion, writers may also provide background information or definition of the research topic. Move 2 directs readers' attention towards a "gap" or niche in the previous studies, which the research aims to fill. Writers usually highlight the gap in the literature or extend the exiting research approach into a new area. In Move 3, writers outline the objectives of the research, showing to readers how the research will fill the gap identified in Move 2.

Language use in the three main moves varies. For example, Move 1 generally employs present or present perfect tense to signal the importance of the research area. Adjectives are also often used to highlight the significance or interest of the topic.

E.g. 1 "The growing importance and popularity of the Internet market in the hotel industry today *have triggered widespread interests* among hoteliers, customers and hospitality researchers to seek an understanding of room pricing behaviour on the Internet so that they can use the knowledge for strategic decision-making" (Mohammed, 2016).

E.g. 2 "Digital composition presents the possibility for digital writers to be both interactive producers and consumers of digital media…. How these digital authors compose in the assorted digital spaces they visit *is an area of interest*…" (Cooperman, 2017).

E.g. 3 "Among these strategies "Green Marketing" and "Environmental Marketing*" have become one of the most prominent* in emerging marketing sector, *attracting a great deal of discussion* among the popular and professional press" (Donikini, 2013).

E.g. 4 "Green Logistics (GL) *has recently received increasing and close attention* from governments and business organisations" (Lin, 2014).

The language in Move 2 is often evaluative as it is through the review of previous studies that writers establish the gap or niche in the area. The gap is often presented as a problem that needs further research and thus, the language of "gap statements" is usually negative in meaning.

E.g. 5 "As far as the existing literature in hospitality is concerned, *limited attempts have been made* to quantify these pricing behaviours" (Mohammed, 2016).

E.g. 6 "*There is little knowledge* regarding how parents can contribute to children's literacy growth with the use of books…*more research is needed* to see what teen mother's perceptions are regarding their influence in their child's language ability to help break this recursive cycle" (Duncan, 2016).

Here are some useful "gap statement" words and phrases (Swales & Feak, 1994, 2004):

Verbs			
disregard	neglect to consider	fail to consider	overestimate
misinterpret	ignore	overlook	is limited to
suffer from	underestimate		
Adjectives			
controversial	questionable	incomplete	inconclusive
misguided	unconvincing	unsatisfactory	
Noun phrase			
Little information/attention/work/data/research			
Few studies/investigations/researchers/attempts			
No studies/data/calculations			
None of these studies/findings/calculations			
Other forms			
However	It remains unclear	It would be of interest to	

As mentioned earlier, Move 3 outlines the purposes of the research. The move tends to employ present tense to give your research current relevance.

E.g. 7 "*The purpose of this study is* to understand how social class background plays a role in student experiences on selective campuses" (Goward, 2018).

E.g. 8 "*This research aims at* developing a content blueprint of a certification programme for infection control nurses in Hong Kong" (Chan, 2012).

E.g. 9 "*This study investigates* the performance of DCSs and conducts design optimization for DCSs and ICSs considering uncertainty of cooling loads and system reliability" (Gang, 2016). (DCS – District cooling system; ICS – Individual cooling system)

Sample introduction

The following introduction was adapted from a PhD thesis in Hotel and Tourism Management, which examines the online pricing behaviour of hotels in Hong Kong. The extract was edited to highlight the three-move framework in an introduction.

Note: […] indicates that parts of the text have been removed.

Introduction	Move
This chapter discusses pertinent issues and concepts that contextualise the study. The chapter commences with a statement of the purpose of the study, which is then followed by the background of the study. The background of the study focuses mainly on describing the evolution of Internet Distribution Channels (IDCs) (interchangeably referred to as online distribution channels) in the hotel industry. The rationale is to provide the motivation for the study and to explain how this platform has created an online market that is revolutionizing hotels' operations and for that matter online pricing of hotel rooms. […]	Advance organiser – provides readers an overview of the Introduction chapter
Over the past several decades, hotels' distribution channels have undergone some significant transformation and evolution (Buhalis & Law, 2008; Choi & Kimes, 2002; Green & Lomanno, 2012; O'Connor, 2008). Specifically, the traditional system of using brochures, guidebooks, travel agencies, tour operators, telephone, fax and call centres as the main distribution channels have expanded to include computerized systems such as the central reservation system, global distribution system, property management system and currently Internet-based systems, which involves hotels' own websites and third-party's such as Online Travel Agencies (OTAs: Buhalis & Law, 2008; Choi & Kimes, 2002; O'Connor, 2008).	Establishing a research territory – providing readers the background information of the research area (i.e. hotel industry)
Remarkably, the introduction of Internet-based systems of distribution is believed to have had the greatest impact on the distribution-channel landscape, thereby establishing itself as a viable and cost-effective system for marketing, selling and distribution of hotel products (O'Connor, 2008). […] In view of the numerous benefits of Internet-based systems of distribution, there has been a significant growth in online reservation of hotel rooms (Green & Lomanno, 2012; Noone & Mattila, 2009). In Noone and Mattila's (2009) study, the authors report that Internet-based reservations for major hotel brands and chains had increased from 27.1% in 2003 to 45.5% in 2007 with similar upward trend expected in the future. A direct consequence of this continuing growth of the online market is that it has considerably increased customers' exposure to hotels' revenue management practices at large and variable pricing in particular (Noone & Mattila, 2009; O'Connor, 2003). That is, due to the adoption of the internet-based systems of distribution, information on room rates that was previously not readily available to customers in the pre-Internet era is now completely transparent and accessible to customers at a click of a mouse. […]	Draw readers' attention to one particular aspect of the research area - internet-based system of distribution

As customers get into the habit of comparing prices and continue to do so effortlessly with the support of price comparison websites, several hospitality scholars are of the belief that the Internet may have an influence on average room rates and the way hotel managers adjust prices over time or implement revenue management (Abrate, et al., 2012; Bitran & Caldentey, 2003; Christodoulidou, Brewer, Feinstein & Bai, 2013; Enz, 2003; Gazzoli, Kim & Palakurthi, 2008; Jayaraman & Baker, 2003; O'Connor, 2008). […]	Introducing the main focus of the study – Online pricing behaviour in the hotel industry
At the outset of the investigations on online pricing behaviour, some researchers (Bakos, 1997; Smith, 2002; Zettlemeyer, 2000) had opined that the Internet market has the ideal characteristics of a competitive market—frictionless, faster, cheaper and better—and for that matter should promote competition, lower average price and eliminate any price differences or price dispersion among sellers of identical goods or services. […] From the studies that have been conducted on online pricing (Borenstein & Rose, 1994; Gaggero & Piga, 2011; Gerardi & Shapiro, 2009; Mantin & Koo, 2009; Obermeyer et al., 2013), there is no explicit definition of online pricing behaviour. However, it can be inferred from the studies of Brynjolfsson and Smith (2000) as well as Baylis and Perloff (2002) that online pricing behaviour is used to refer to observable differences and patterns in prices of identical goods that are measurable in terms of central tendencies and/or dispersions. Consistent with these scholars, this study uses the term "online pricing behaviour" to refer to quantifiable outcomes in the price setting of hotel rooms. More specifically it refers to the frequency of price change, direction or pattern of price change and magnitude of dynamic price dispersion.	Introducing and defining the term "online pricing behaviour"
As far as the existing literature in hospitality is concerned, limited attempts have been made to quantify these pricing behaviours. In a study conducted by Abrate et al. (2012), the authors provided evidence to indicate that dynamic price dispersion was substantial in European hotels' market but did not analyse the factors contributing to the magnitude of the dynamic price dispersion. The authors' concentration was on describing the dynamic pricing structure of hotels which they found to vary according to type of customer (weekday booking or weekend booking), star rating of hotels and number of hotels with available rooms in a city. […]	Indicating a gap
Empirical evidence regarding the frequency of price change on the Internet has only been provided in the markets for durable goods (Brynjolfsson & Smith, 2000) and the airline industry (Gillen & Mantin, 2009; Mantin & Gillen, 2011) but not in the hotel industry. In the light of the incomprehensive literature on online pricing behaviour in the hotel industry, this study focuses further attention on examining online pricing data to contribute to the emergent literature on online pricing behavior of hotels.	Indicating a gap Occupying a niche – outline the purpose of the study
Subsequent to the transparency of room rates and customers' exposure to	Showing the importance of the research topic

revenue management pricing, some hospitality researchers have underscored the importance of customers' familiarity and knowledge of pricing behaviour to the successful implementation of revenue management and long-term viability of the Internet market. For example, in Noone and Mattila's (2009) investigation of the effect of Internet price presentation strategies on customers' willingness to book a hotel room, they found that customers' familiarity with revenue management pricing practices improves customers' willingness to book online. […] While consumer familiarity and knowledge of pricing behaviour have been documented to be important for the successful implementation of revenue management and promotion of long-term viability of the Internet market, there is surprisingly limited research on online pricing behaviour in the hotel industry as compared to the airline industry, where extensive research has been conducted (Borenstein & Rose, 1994; Gaggero & Piga, 2011; Gerardi & Shapiro, 2009; Mantin & Koo, 2009; Obermeyer et al., 2013). Meanwhile, the equivalent studies in the airline industry have demonstrated the value of these studies in contributing to identify the sources of price advantage for strategic consumers' decision making. […] Thus, for a subject of this importance, which has rather received limited attention in the hotel industry, it is appropriate to focus more attention on characterizing online pricing behaviour beyond the current discussion about dynamic pricing structure, which focuses mainly on describing price path to examining the factors that influence the different aspects of pricing behaviour. The present study contributes to this research agenda by addressing the following specific gaps.

Indicating a gap

Highlighting the significance of the research

First, it is widely acknowledged that the Internet enables hotel operators to frequently adjust room rates due to the implementation of revenue management software and the relatively low cost of changing prices on the Internet. However, to date, there is yet to be an empirical study in the hospitality literature that systematically investigates the frequency of price change over a given period of time and to examine the factors influencing the frequency of price adjustment.

Second, the lack of studies that examine the predictors of price change limits the knowledge of strategic consumers and competitors who may wish to predict, at least, the direction of future changes in room rates based on observable characteristics of hotels and market structure.

Last but not least, empirical observation and literature on online pricing data suggests that room rates vary dynamically over booking histories due to the implementation of revenue management (Abrate et al., 2012; Bitran & Caldentey, 2003). However, little is known about the magnitude of dynamic price dispersion of different hotel segments. Moreover, there is the absence of empirical study to link the magnitude of dynamic price dispersion to location attributes of hotels and localized competition or market structure within which hotels operate. Especially, the effects of localized competition and spatial heterogeneities on dynamic price dispersion have not been investigated even though economic theory (i.e. the structure-conduct-performance theory) suggests there could be causal linkages. The closest study to this area in the hospitality literature is the one provided by

Detailed description of the research gaps

Abrate et al. (2012), in which the authors link the structure of dynamic pricing (but not the magnitude of dynamic price dispersion) to the type of customer (weekday or weekend bookings), hotels' star rating and the number of hotels with available room during a booking period. […] In line with the stated research problem and the emergent research questions, the broad objective of this study is to analyse online price behaviour of hotels in Hong Kong to determine the frequency of price change, the patterns of change and the extent of dynamic price dispersion, and to identify the factors influencing these price-setting behaviours. Specifically, the study seeks to: 1. Measure the frequency of price change online and analyse the factors influencing it in the Hong Kong hotel market; 2. Determine the patterns of price change online and examine the influencing factors in Hong Kong hotel market; and 3. Quantify the size of dynamic price dispersion and identify the factors that determine its magnitude in Hong Kong hotel market. To achieve the above objectives, a conceptual framework for this study is established on three major predictors: market structure characteristics; hotel characteristics and location attributes. Source: Mohammed, I. (2015). *An analysis of online pricing behaviour of hotels in Hong Kong.* (Doctoral thesis, The Hong Kong Polytechnic University).	Occupying the niche – presenting the objectives of the study and showing how the study addresses the aforementioned research gaps

Activity 1

Read the following extract from the Introduction chapter of a PhD thesis, which examines the effectiveness of writing conferences.

i) Based on what you read, discuss with your group mates:
   a) How did the writer introduce the research topic and establish its significance?
   b) Where did the writer evaluate previous research and indicate specific research gaps? In which paragraph(s)? Give examples to support your answer.
   c) How does the writer's study address the research gap?
   d) Will the study make any contribution to the field? In which paragraph(s), did the writer mention the contribution?

ii) In the right hand column, write what the writer has achieved in each paragraph. You can also work in pairs and conduct a sentence-by-sentence analysis on the sample introduction.

	Introduction	Move
Para. 1	[1]Tutoring writing is a routine activity that teachers around the globe perform with students on a daily basis. [2]Many teachers and academic scholars believe in the practice of providing oral feedback on student writing and that a productive channel of feedback provision is one-to-one writing conferences (e.g. Freedman and Sperling, 1985; Nystrand and Brandt, 1989; Sperling, 1991). [3]This face-to-face feedback mode has gained popularity in recent decades, and it is postulated that its dialogic nature allows student participation in the critical evaluation of text, and encourages student-teacher interaction. [4]But are writing conferences as helpful as people say they are?	
Para. 2	[5]The talk in the situated learning context of a writing conference is believed by many scholars to encourage students to articulate problems, express intentions, exchange ideas with the teacher, and evaluate the text (e.g. Zamel, 1985; Leki, 1992; Arndt, 1993). [6]The social processes embedded in the conversation are also said to allow teachers to understand students' thinking and writing difficulties, and to design pedagogical strategies as they listen to student concerns (Walker 1992; Reesor, 2002). [7]Some researchers, however, have queried the claimed benefits of the conference, and whether conferencing really provides opportunities for student initiation and expression (e.g. Ulichney and Watson-Gegeo, 1985; Johnson, 1993;	

Newkirk, 1995; Black 1998). [8]More observations of conversational behaviour are therefore necessary to find out how the social dimensions of the writing conference are achieved, and whether the assumptions that conferencing is an efficient channel for commentary and collaboration are true.

……

Para. 3

[9]With the increase in the popularity of writing conferences …, a number of resources in the form of strategies or guidebooks have been written on conference strategies, e.g. Garrison (1981), Reigstad and McAndrew (1984), Murray (1985), Clark (1985), Harris (1986), Newkirk (1989), Phenix (1990), McAndrew and Reigstad (2001), some of which contain lists of 'dos and don'ts'. [10]Much of the advice in these resources seems to be based on personal experience rather than on research evidence. [11]Little empirical research on conferencing in classroom contexts has been conducted; and being qualitative case studies, they covered a small number of subjects only. [12]The aspects of conferencing that have been examined are interesting but limited, such as participant attitudes (Carnicelli, 1980; Zamel, 1985), conference discourse (Freedman and Katz, 1987), the effects of conferencing on writing (Goldstein and Conrad, 1990; Patthey-Chavez and Ferris, 1997), and verbal interactions (Thonus, 2002).

Para. 4

[13]Embedded in face-to-face conferencing are complex issues and factors that intertwine and mutually affect one another. [14]While certain verbal conferencing activities, such as topic-initiation and turn-taking, have been singled out for examination, studies that focus on the participants or explore their interactions holistically have so far been scanty. [15]Although a vast amount of literature on human communication stresses the importance of nonverbal behaviour, few studies of writing conferences have investigated the influences of nonverbal exchanges on the conference and its interactants. [16]The literature on writing conferences also shows a lack of comprehensiveness in that previous studies did not explore the whole picture of conferencing in terms of what happens *before* conferences, i.e. the kinds of preparation and their influences on conferences; and only a limited number of studies have investigated the perceptions of conference

interactions by *both* the teacher and student participants rather than by one of the parties only.

Para. 5

[17]A review of the literature further reveals conflicting views of the writing conference with regard to such matters as the roles and responsibilities of the participants, the focus of the teacher, turn-taking and idea exchange, and teacher-guided or student-guided agendas. [18]This inconclusiveness has led to doubts about the value of conferencing as a pedagogical tool (Black, 1998).

……

Para. 6

[19]To investigate the complexities of the face-to-face writing conference that include an examination of not only verbal exchanges but also of nonverbal interactions, and an exploration of not only the features of the encounter but also the feelings therein and thereafter, there is the need to adopt a comprehensive analytical framework that observes not just the conference *activities* (e.g. turn-taking, interruptions) but also the *actors,* i.e. the participants, including their beliefs, expectations and perceptions of what happens in conference interactions.

Para. 7

[20]In view of the above, this study sets out to examine, with a breadth and a depth that have seldom been applied in previous research, the writing conferences of four teachers and eight students in a naturalistic university classroom setting in Hong Kong. [21]Its aims are to explore the nature and dynamics of these L2 writing conferences, as well as the effects of the verbal and nonverbal interactions within them, in the hope of resolving some of the conflicting findings of previous research. [22]Following a qualitative case study method, and employing an approach that seeks to "view situations as they appear to those directly involved in them" and to "appreciate how and why actors perceive things in the ways they do" (Cuff, Sharrock, and Francis, 1990, pp.151-2, emphasis in original), this study presents a holistic investigation of the verbal and nonverbal interactions that were video-recorded during the conferences, and the contexts in which these exchanges took place. [23]The data collection, was however, not restricted to the conferences themselves, but also included the pre-conference stage of expectation and preparation and the post-conference stage of interviews and video stimulated

	recalls. [24]Detailed transcriptions of the videos allowed participant behaviour to be systematically categorised and coded in a search for analytic foci, patterns and unique features. [25]Together with the interview data, they suggest reasons why some conferences are more highly valued than others.	
Para. 8	[26]Analysis of my data has led me to query the use in the existing literature of the notions of 'success' or 'effectiveness' to describe conferences, particularly regarding their relationship with draft revision. [27]Since revision can depend on many variables, this study suggests that a conference may be best judged by the affect and perceptions of its interactants. [28]I therefore in the later chapters of this report propose a new analogy with health to discuss the holistic considerations of the dynamic process of interactions in teacher-student conferences, explore the fresh concept of interaction spaces in the interplay of conference parameters, and conclude with a description of a number of dynamic balances that are pertinent to healthy conferences.	
Para. 9	[29]The chapters of the dissertation are organised as follows: Chapter 1 Introduction: the current chapter. Chapter 2 Literature Review: this …… and presents limitations and gaps in earlier research. Chapter 3 Research Questions and Methodology: this states …	

iii) In the sample introduction, identify the phrases that:

Show a belief or a positive act	Show a disbelief or a doubt
E.g. Many teachers and academic scholars believe in …(sentence 2, para. 1)	
Indicate a gap	Indicate inconsistencies in previous research

You may use the following verbs and phrases to **indicate inadequacies in previous research**:

i) Verbs

- Previous studies have underestimated
  overestimated
  ignored
  neglected
  failed to consider /include/take into account / take account of
  overlooked

ii) Noun phrases

- The limitations / restrictions of previous studies …
- Little research has been conducted in…
- Little attention has been given / devoted to…
- … has received little attention.
- Only a limited number of studies / researchers / investigations have …
- Few studies / researchers / investigations have …
- A limited body of research has been conducted …
- Hardly any study has been conducted …
- Hardly any data has been collected …
- There is still much to be learned…
- There is still a need to clarify / investigate …
- There remains a gap in our knowledge / understanding of … to be filled.
- Our knowledge / understanding of … is incomplete / far from complete.

iii) Adjectives

- Studies that focus on … have so far been scanty;
- Y's results are found to be unconvincing / incomplete
- Previous findings/results are debatable
  controversial
  questionable
  inconclusive
  open to question
  dated

iv) Other phrases

- The findings/results of previous studies *show inconsistencies / a lack of agreement*
- X's (2018) findings *have been refuted by* Y (2019) …
- Y's (2019) findings *do not match those of* X's (2018)…
- X's (1985) study was conducted over three decades ago using research instruments that are *no longer considered to be* accurate.

III. Language use in Introduction

This section presents the types of language use in the Introduction chapter of a thesis: i) writing definitions; and ii) language of comparison and evaluative language.

i) Writing definitions

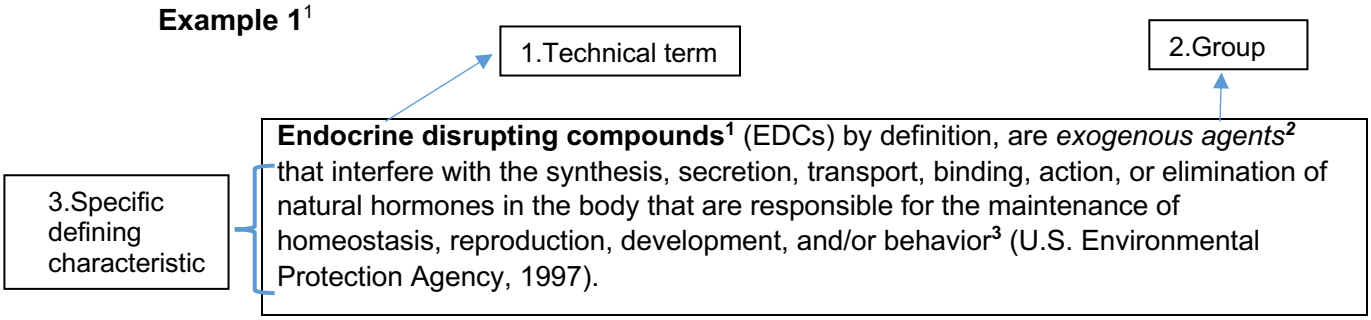
As shown in Section II (Moves in an introduction), one sub-move in Move 1 is defining terms. In the Introduction chapter, writers may need to define specific terminology that readers may not be familiar with. While providing a definition to the technical terms, writers can also demonstrate to readers (e.g. examiners) their specialist knowledge of the research topic.

In general, a definition has the structure shown below (Swales & Feak, 2004, p. 56):

(A) Term is (a) Class Wh-word Distinguishing Detail

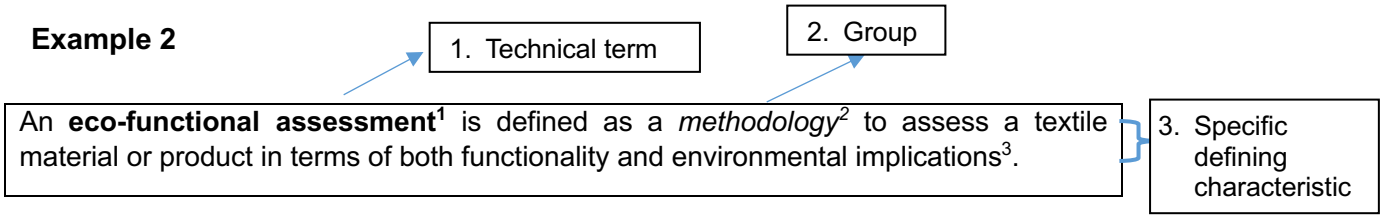
A solar cell is a device that/which converts the energy of sunlight into electric energy.

Example 1[1]



Adapted from: L. Xu. (2014) *Degradation of Refractory Contaminants in Water by Chemical-Free Radicals Generated by Ultrasound and UV Radiation.* (Ph. D dissertation), Dept. Civil and Env. Eng., HKPOLYU, Hong Kong. [Online] Available: HK Poly U Electronic Theses.

Example 2



Adapted from Muthu, S. S. (2012). *Eco-functional assessment of grocery shopping bags.* Doctoral thesis. Hong Kong Polytechnic University

In the following example, the writer provides a classification of a specific type of material.

Example 3

1. Material 2. REDOAs, OSOAs, Raman amplifiers and OPA are classifications of the material.

Optical amplifiers[1] can be **classified into** rare-earth (RE) doped optical amplifiers **(REDOAs)**, semiconductor optical amplifiers **(SOAs)**, **Raman amplifiers** and optical parametric amplifiers **(OPA)** [2].

Adapted from: B. Chen. (2015). *Rare Earth Ions Doped Oxide Glass Materials.* Ph. D dissertation, Dept. of Elect. Eng., CITYU, Hong Kong. [Online] Available: CityU Institutional Repository.

[1] Examples 1, 3 and 5 and Activity 2 in this handout were adapted from the learning materials developed by OSAWEC.

ii) Language of comparison and evaluative language

While providing the background information of a research topic, writers often review relevant literature to show what studies have been conducted in the area and such review usually includes a comparison among previous studies. Writers also evaluate these studies to highlight their strengths or weaknesses. The samples below show a range of language of comparison and evaluative language used by writers. […] indicates text has been edited due to space constraints.

Example 4

The following text was extracted from the Introduction chapter of a doctoral thesis in Computing. In the extract, the writer evaluated the software engineering methodologies that have been adopted in the area and commented on their ineffectiveness.

Key: **Language of comparison** ***Evaluative language***

Furthermore, despite being compliant with international standards for the development of safety-critical CST systems is widely mandated, the software requirements have been repeatedly recognized to be the most ***problematic*** within the software development lifecycle (Lamsweerde, 2000). These problems are primarily due to the fact that although what is being developed is **highly** ***interactive*** software with ***significant*** HCI components, most software engineering methodologies offer no mechanisms to explicitly and empirically identify and specify user needs and usability requirements. Even though most software projects start with some form of requirements analysis and specifications, but the initial requirements are not done well because those involved have ***inadequate*** understanding of the human factors that should be ***paramount*** in requirement analysis (Diaper & Sanger, 2006). They also ***fail*** to test and validate requirements with end-users before, during and after the development. **On the other hand**, analysis methods offered by HCI experts and existing software engineering practice are often ***discordant***. The report of Stone Man version of SWEBOK (Software Engineering Body of Knowledge) [SWEBOK, 2000] lists HCI and related knowledge as "related disciplines" in software engineering field […] As a result, the developed systems generally meet all functional requirements, and yet are not ***effective, efficient and satisfactory*** for use. The frequent situation where large numbers of requests to modify are made after the systems are deployed (Seffah & Gulliksen, 2005) can be attributed to such ***inadequate*** methodologies.

(CST systems - Complex socio-technical systems; HCI - Human computer interaction)

Adapted from Tung, Y. W. (2012). An approach to the usability evaluation of the human-computer interaction of a heterogeneous safety-critical complex socio-technical system. Doctoral thesis, Polytechnic University of Hong Kong.

Example 5

The following example was extracted from a thesis in Electrical Engineering. In this example, REDOAs are positively compared with SOAs and Raman amplifiers. The negative aspects of SOAs are presented first and later, Raman amplifiers are negatively evaluated.

Key: **Language of comparison** ***Evaluative language***

Among these optical amplifiers, rare-earth doped optical amplifiers (REDOA)s **are** ***advantageous*** **over** the semi-conductor optical amplifiers (SOAs) and Raman amplifiers. **Compared with** REDOA, SOA ***is*** ***disadvantageous*** because it involves the interference between adjacent pulses in the saturation regime as a result of the short gain recovery time. SOAs produce a ***significant*** pulse-pattern distortion in the output signal at 25 Gbs, **whereas** no ***significant*** degradation over 100 Gps has been observed in REDOAs. In addition, the coupling loss between an SOA and an optical fiber **is much** ***larger*** **than that between** REDOAs and optical fibers. **On the other hand**, the ***major drawback*** of a Raman amplifier **compared with** an REDOA is its ***low efficiency***. Raman amplifiers have **relatively** ***poor*** pumping efficiency at low signal powers resulting in **much** ***easier*** gain clamping.

Adapted from: B. Chen. (2015). *Rare Earth Ions Doped Oxide Glass Materials*. (Ph. D dissertation), Dept. of Elect. Eng., CITYU, Hong Kong. [Online] Available: CityU Institutional Repository.

Activity 2

The text below was extracted from the Introduction chapter of a Computer Science thesis. Can you identify the following language features in the extract?

- Language of definition
- Language of comparison and contrast
- Evaluative language

Note: […] indicates text has been edited due to space constraints.

[…] Taking a utilitarian criterion, we identify generic summaries and query-focused summaries. Generic summaries do not address a particular user need whereas query-focused summaries are produced in response to a user need or query. The last criterion is the number of source documents, which can distinguish single-document summarization from multi-document summarization. […]

By definition, text summarization is “the process of distilling the most important information from a text to produce an abridged version for a particular task and user” (Mani and Maybury, 1999; as cited in Zhang, 2012, p. 2). It was born out of the need for an efficient way to obtain the most important information from a large body of documents without reading all of them. With a history

of over 50 years, it proves to be one of the most vigorously explored frontiers in NLP applications. What was once believed to be a human-privileged creative task is now extensively automated in modern computing labs and commercial packages. […] In the growing community of text summarization, research efforts are traditionally lopsided towards the information contained in summaries, not the presentation of the information, although the latter is arguably no less important. This dissertation is exclusively devoted to one crucial aspect of summary presentation –coherence.

Adapted from: R. Zhang. (2012). *Coherence-targeted Text Summarization*. (Ph. D dissertation). Dept. of Comp. Eng., HKPOLYU, Hong Kong. [Online]. Available: HK PolyU Electronic Theses.

iii) Verb tenses

In the Introduction, a combination of verb tenses is used.

Content	Language feature	Example
Research topic/ Problem description	• Present tense To refer to a process, a fact or information established as a fact in the field. • Present perfect tense To provide information or facts to create a background for the topic • Present continuous tense To refer to a currently developing, ongoing situation or statements that have current relevance (i.e. important now), where the emphasis is on the present time. It is different from the present perfect tense which details an unspecified period of time.	*E.g. 1 Build products are rarely defect-free as quality control is less efficient than in a factory environment.* *E.g. 2 Development of building energy efficiency standards has been an important issue in Hong Kong and mainland China in the past two decades.* *E.g. 3 Reanalysis models are rapidly gaining popularity for simulating wind power output due to their convenience and global coverage. However, they should only be relied upon once thoroughly proven. This paper reports…*
Research gap	• Present tense **OR** present perfect tense • Expressions that indicate causality, concession, comparison (e.g. *hence*, *however, although* and *then*)	*E.g. 4 Life cycle is an important issue that is well documented in the literature… However, few studies consider life cycle assessment (LCA) and life cycle costing (LCC) for high-rise precast buildings located in dense urban environments.* *E.g. 5 Although studies have been done on energy use and GHG emissions in the life cycle of buildings, very few have focused on the construction stage in particular and none comprehensively.*
Objectives	• Present tense (dissertation-oriented, i.e. it views research as the dissertation that the readers currently hold in their hands) • Past tense (research-oriented, i.e. it	*E.g. 6 …this study investigates current technologies adopted in precast housing construction and the challenges confronted in the process.*

	views research as a completed study)	*E.g. 7 The objectives of this research were to develop a numerical wind tunnel model with a self-developed solver on an open source CFD platform, to validate the model by self-conducted experimental measurement…*
Overview of the major sessions	• Present tense **OR** future tense	*E.g. 8 The rest of this paper is organized as follows. The next section introduces the input-oriented Malmquist productivity index. An empirical study in China from 2004 to 2009 is then presented and discussed. Conclusions are given in the last section.* *E.g. 9 The paper will first study and compare both the BASIX and BEAM Plus assessment systems. Second, the advantages and pitfalls of the mandatory and voluntary approaches will be identified and discussed.*

E.g. 1, 4 Jaillon, L. and Poon, C-S. (2010). Design issues of using prefabrication in Hong Kong building construction. *Construction Management and Economics,* 28, 1025-1042.

E.g. 2 Hui, S. C. M. (2000). *Building energy efficiency standards in Hong Kong and mainland China.* The Proceedings of the 2000 ACEEE Summer Study on Energy Efficiency in Buildings, Pacific Grove, California, 20-25 August. Available at: http://aceee.org/files/proceedings/2000/data/index.htm [Accessed 31 August 2016].

E.g. 3 Staffell, I., & Pfenninger, S. (2016). Using bias-corrected reanalysis to simulate current and future wind power output. *Energy*, *114*, 1224-1239.

E.g. 5 Yan, H., Shen, Q., Fan, L. C. H., Wang, Y. and Zhang, L. (2010). Greenhouse gas emissions in building construction: A case study of One Peking in Hong Kong. *Building and Environment, 45*(4), 949-955.

E.g. 6 - Li, C. Z., Xue, F. and Shen, G. Q. (2015). *Building information technologies and challenges in precast housing construction in Hong Kong.* Paper presented at the 7th International Conference of SuDBE2015, Reading, UK, 27-29 July. Available at: <http://www.sudbe.com/ueditor/php/upload/file/20160520/1463743981517758.pdf> [Accessed 31 August 2016].

E.g. 7 - Zhang, Y., Kwok, K. C. S., Liu, X. –P. and Niu, J. –L. (2015). Characteristics of air pollutant dispersion around a high-rise building. *Environmental Pollution,* 204, 280-288.

E.g. 8 - Xue, X., Wu, H., Zhang, X., D, J. and Su, C. (2015). Measuring energy consumption efficiency of the construction industry: The case of China. *Journal of Cleaner Production,* 107, 509-515.

E.g. 9 - Ho, D. C. W., Ge, J. X. and Liusman, E. (2013). *The mandatory and voluntary approaches to sustainability: BASIX vs BEAM Plus.* The Proceedings of the 19th CIB World Building Congress (WBC13), Brisbane, Australia, 5-9 May. Available at: <http://hub.hku.hk/bitstream/10722/187304/1/Content.pdf> [Accessed 31 August 2016].

IV. Research aim, objectives and questions

One crucial task in any research studies is defining its aim, objectives or questions. What is the main purpose of your research study? What objectives or questions does your research address? It is important that you formulate the aim and objectives of your research in an appropriate manner as they determine the scope and direction of your study. The three terms "research aim", "research objectives", and "research questions" have specific meanings. You need to understand their meaning in order to apply them appropriately in your thesis.

Research aim

A research aim refers to the overarching purpose of a research study. It provides readers with clarity as to what your research is about. Research aims are usually clear and brief statements that are positioned in the Introduction chapter of a research thesis. They tend to be broad rather than specific, indicating clearly what your research sets out to accomplish. An example is:

Background information

Due to the spread of COVID-19 worldwide, a large number of universities had to close their campuses. To maintain teaching and learning during this disruption to the traditional teaching, most universities have adopted online teaching model. The current study aimed at investigating the efficacy of various online teaching modes as well as comparing a proposed combined model of online and flipped learning to other online and traditional models.

(Tang et. al, 2020, p. 1)

Research aim

While research aims indicate **what** your research intends to achieve, research objectives specify **how** your research will achieve the aims.

Research objectives

Research objectives state in more detail the specific issues that your research study plans to investigate. A research aim is usually followed by at least two or three research objectives, each of which represents a major section of your research study. Research objectives are often presented in a numbered list as follows:

Objective 1: To examine students' perceptions about online teaching during the COVID-19 situation.
Objective 2: To compare a combined model of online learning and flipped classroom with traditional and online teaching.

Research questions

In some cases, researchers state their research objectives in the form of questions. The aforementioned research objectives are rephrased as research questions:

Research question 1:
What are students' perceptions about online teaching during the COVID-19 situation?

Research question 2:
How does a combined model of online teaching and flipped classroom compare to traditional and online teaching?

(Tang et. al, 2020, p. 1)

Table 1 below shows the definition of the three commonly used terms in thesis writing

Table 1: Definition of the three terms

Term	Definition
Research aim	A statement indicating the general aim or purpose of a research study. A research study usually has one broad aim.
Research objectives	Specific statements indicating the key issues to be focused on in a research study. A research study usually has several specific research objectives.
Research questions	An alternative way to present research objectives – the key issues to be focused on in a research study are presented in the form of questions.

Adapted from Thomas and Hodges (2010, p. 39)

Here are some online resources that can help you understand more about these three commonly used terms:

- Differences between aims and objectives: https://learn.solent.ac.uk/mod/book/view.php?id=116233&chapterid=15294
- An example showing the use of "aims", "objectives" and "research questions" https://www.ncbi.nlm.nih.gov/books/NBK299448/
- A video clip showing the definition and examples of "research aim, objectives and questions" https://www.youtube.com/watch?v=_bD9CvNEL-c

V. Guiding questions

The goal of writing the Introduction chapter is to show that your study is a well-conceived initiative that provides a promising solution to a crucially important need or an issue of significance. You may consider the following questions when writing the chapter:

Establish necessity

- What is the issue/need that needs to be addressed?
- What is the basis of the observation that this is indeed an issue/need?
- Why is it crucially important for your study to be conducted?

Propose a solution

- What does your study set out to do?
- How does it relate to the issue/need identified?
- Why does it represent a feasible and sustainable solution to the problem?
- Is there any theoretical and/or empirical basis for the solution?
- How does it make use of existing knowledge and prior experience?
- Why is your proposed study superior to alternative solutions?

Establish value

- What are the specific objectives of your study?
- What kinds of impact may be expected?
- Is the impact significant enough? Will the impact last?
- Will it provide basis for future development/research?

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